

$$\begin{aligned} 5 \quad (a) \quad 5000 \text{ revolutions per minute} &= \frac{5000}{60} \text{ revolutions per second} \\ &= 83.33 \text{ revolution per second} \end{aligned}$$

$$\begin{aligned} \text{Angular velocity } \omega &= 2\pi f \\ &= 2\pi (83.33) \\ &= 524 \text{ rad s}^{-1} \end{aligned}$$

$$\begin{aligned} (b) \quad (i) \quad \text{The period of the Earth's rotation about its axis is} \\ T &= 24(60)(60) \\ &= 86,400 \text{ s} \end{aligned}$$

$$\begin{aligned} \text{Angular velocity } \omega &= \frac{2\pi}{86400} \\ &= 7.27 \times 10^{-5} \text{ rad s}^{-1} \end{aligned}$$

$$\begin{aligned} (ii) \quad \text{Centripetal acceleration, } a &= r\omega^2 \\ &= 6.38 \times 10^6 (7.27 \times 10^{-5})^2 \\ &= 3.37 \times 10^{-2} \text{ m s}^{-2} \end{aligned}$$

- (c) (i) Acceleration of free fall of a point on Earth depends on two factors: radius of Earth at the point and the radius of that point's circular path.

Earth is not a perfect sphere. The distance from the centre of the Earth to the Equator is larger than the distance to the poles. Hence the acceleration of free fall at the equator is smaller than that at the poles.

The equator undergoes circular motion about the Earth's axis, whereas the poles do not. At the equator, part of the gravitational force has to provide for the centripetal acceleration hence the acceleration of free fall at equator is lower.

- (ii) The difference in Earth's radius at the poles and Equator is very small compared to the average radius of Earth.

The Earth's centripetal acceleration (0.0337 m s^{-2} from part (b)(ii)) is negligible compared to the value of 9.81 m s^{-2} .

- (d) (i) Newton's law of gravitation states that two bodies attract each other with a gravitational force which is directly proportional to the product of the two masses and inversely proportional to the square of the distance between the centres of the two masses.

- (ii) Consider a point mass m located at the surface of the Earth. The force of attraction between the Earth and the mass m is $F = \frac{GMm}{R^2}$

The gravitational field strength g at a point is the gravitational force F acting per unit mass placed at that point.

$$g = \frac{F}{m}$$

$$g = \frac{1}{m} \left(\frac{GMm}{R^2} \right)$$

$$g = \frac{GM}{R^2}$$

- (e) (i) From (d)(ii), on Earth's surface, gravitational force on the satellite of mass m is:

$$mg = \frac{GMm}{R^2} \dots\dots\dots (1)$$

At a distance r from the centre of Earth, the gravitational force on the same satellite provides the centripetal force on the satellite:

$$mr\omega^2 = \frac{GMm}{r^2} \dots\dots\dots (2)$$

$$\begin{aligned} (1) \text{ divide by } (2), \quad r &= \left(\frac{gR^2}{\omega^2} \right)^{1/3} \\ &= \left[\frac{(9.81)(6.38 \times 10^6)^2}{(7.27 \times 10^{-5})^2} \right]^{1/3} \\ &= 4.23 \times 10^7 \text{ m} \end{aligned}$$

Note: Some students may be unable to substitute for the mass of the Earth by using the expression derived in (d)(ii).

- (ii)
 - 1. The satellite must lie on a plane containing the Earth's equator.
 - 2. The satellite rotates from west to east, following the Earth's direction of rotation.

(f) Polar advantage 1: Satellites in low orbits are closer to the Earth hence the delay between transmission and receipt of signals is reduced.

Polar advantage 2: Instead of being limited to the Equator, polar orbit satellites can fly over and serve every part of the world.

Geostationary advantage 1: Since the satellites are always at the same spot above the equator, continuous transmission between the ground station and the satellites is possible.

Geostationary advantage 2: During launch, Earth's west-east rotation provides